

Ethanol Blending in Gasoline – Bulgaria

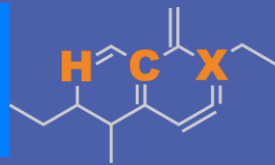
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**U.S. GRAINS &
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Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Bulgaria



In 2024, gasoline consumption in Bulgaria was 2,342 million liters (619 million gallons) and growth rate was -2.3%. Gasoline production and net imports were 2,199 and 143 million liters (581 and 38 million gallons) correspondingly. Bulgaria complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.5 RON.

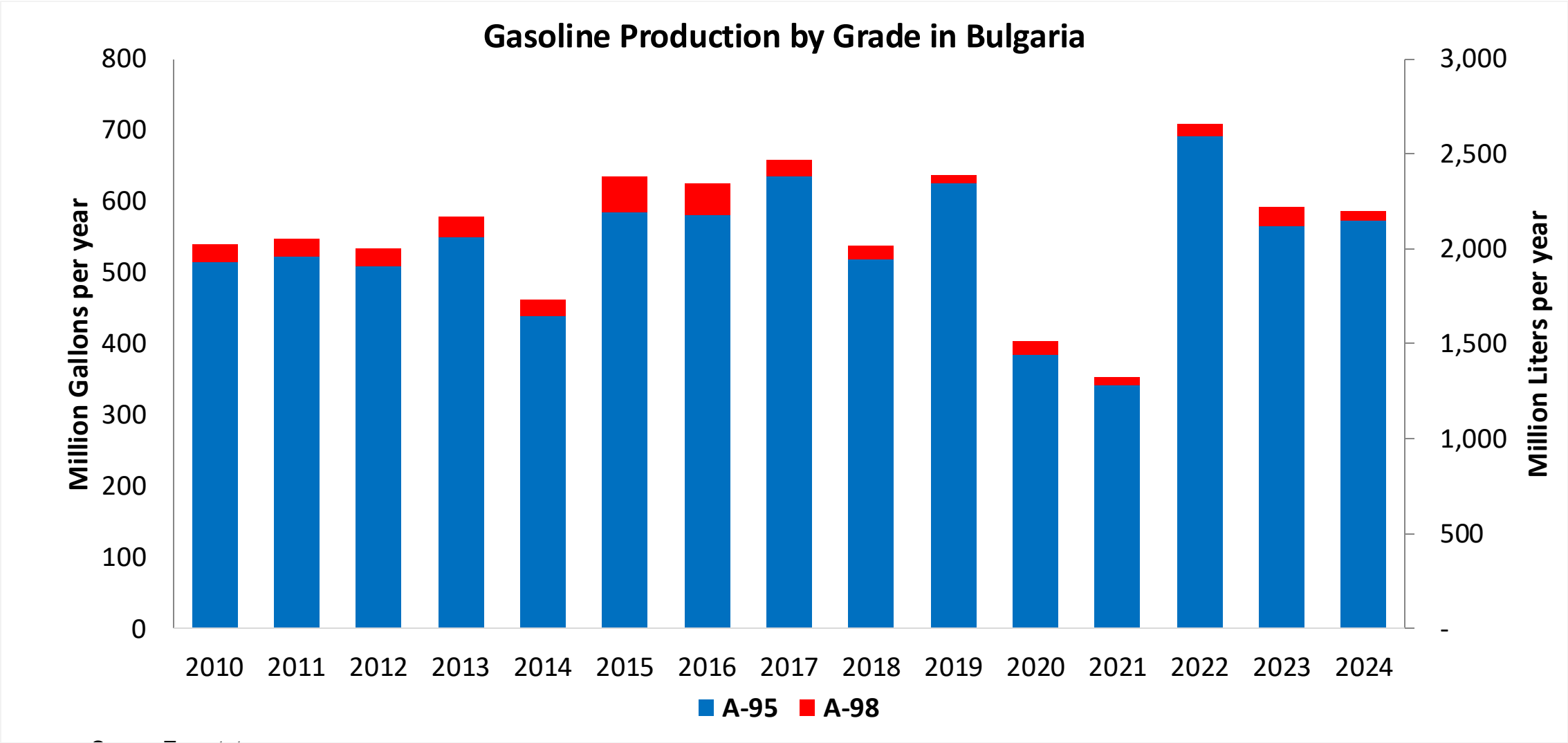
In 2024, ethanol consumption in Bulgaria was 54 million liters (14 million gallons) representing 2.3% of gasoline demand. Ethanol production and Net Imports were 40 and 14 million liters (14 and 4 million gallons) correspondingly. Ethanol sources are Corn 45% and Cereals 55.

Source: Eurostat, European Commission

The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexadiene. The structure features a six-membered ring with two double bonds. Substituents include a methyl group, a propyl group, and a group labeled 'X'. The letters 'H', 'C', and 'X' are highlighted in orange in the original image.

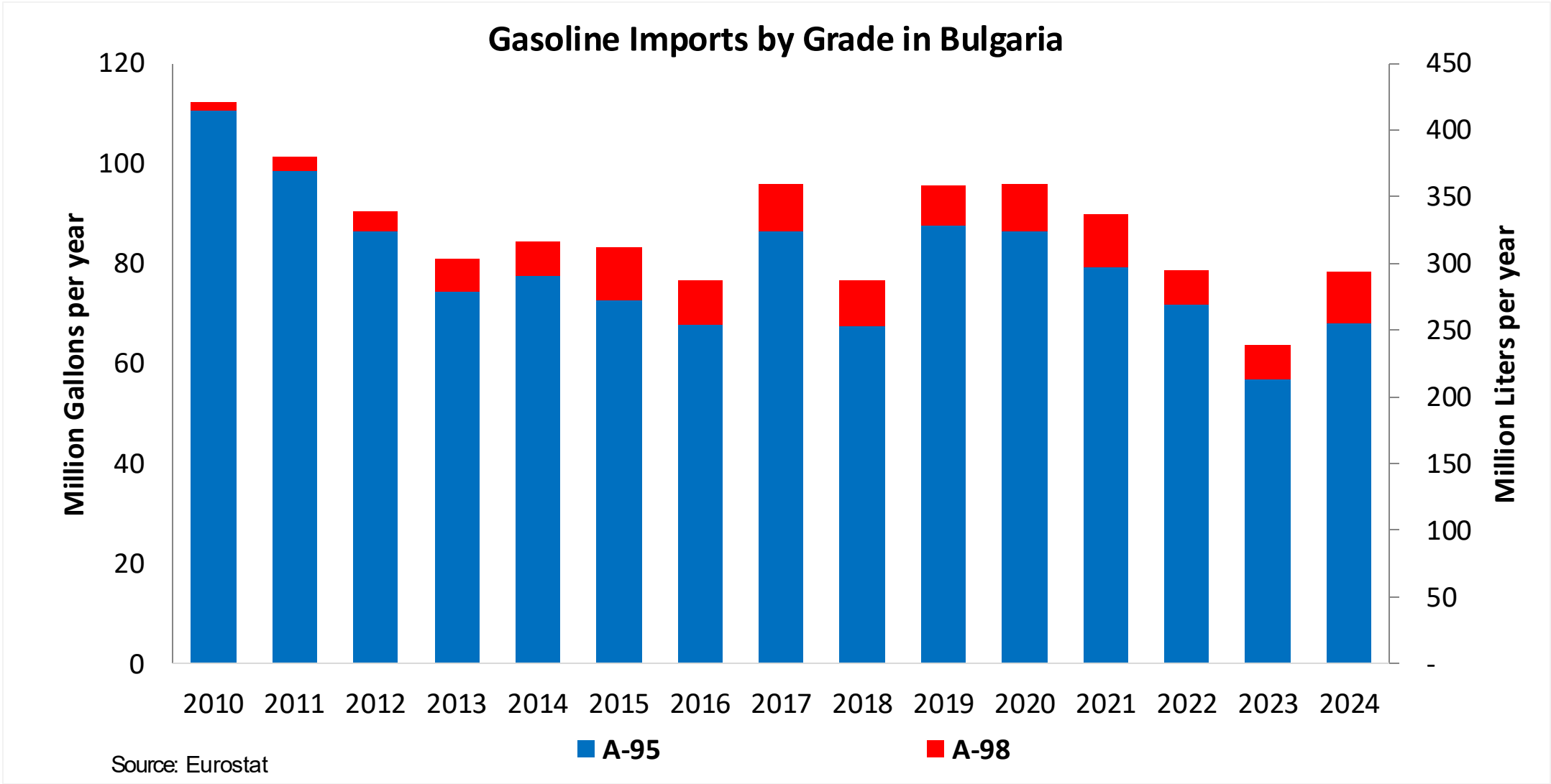


Gasoline Production in Bulgaria

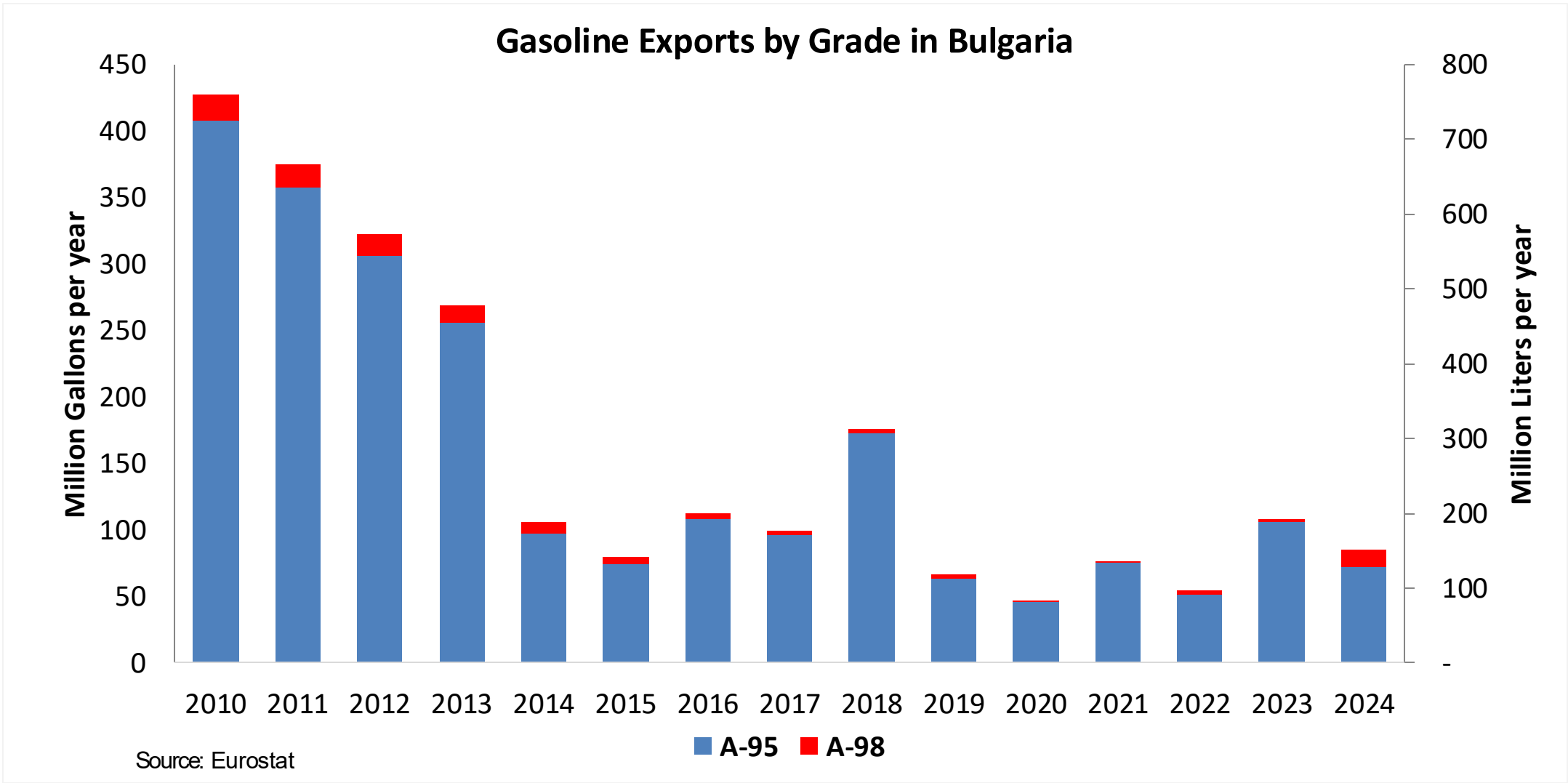


Source: Eurostat

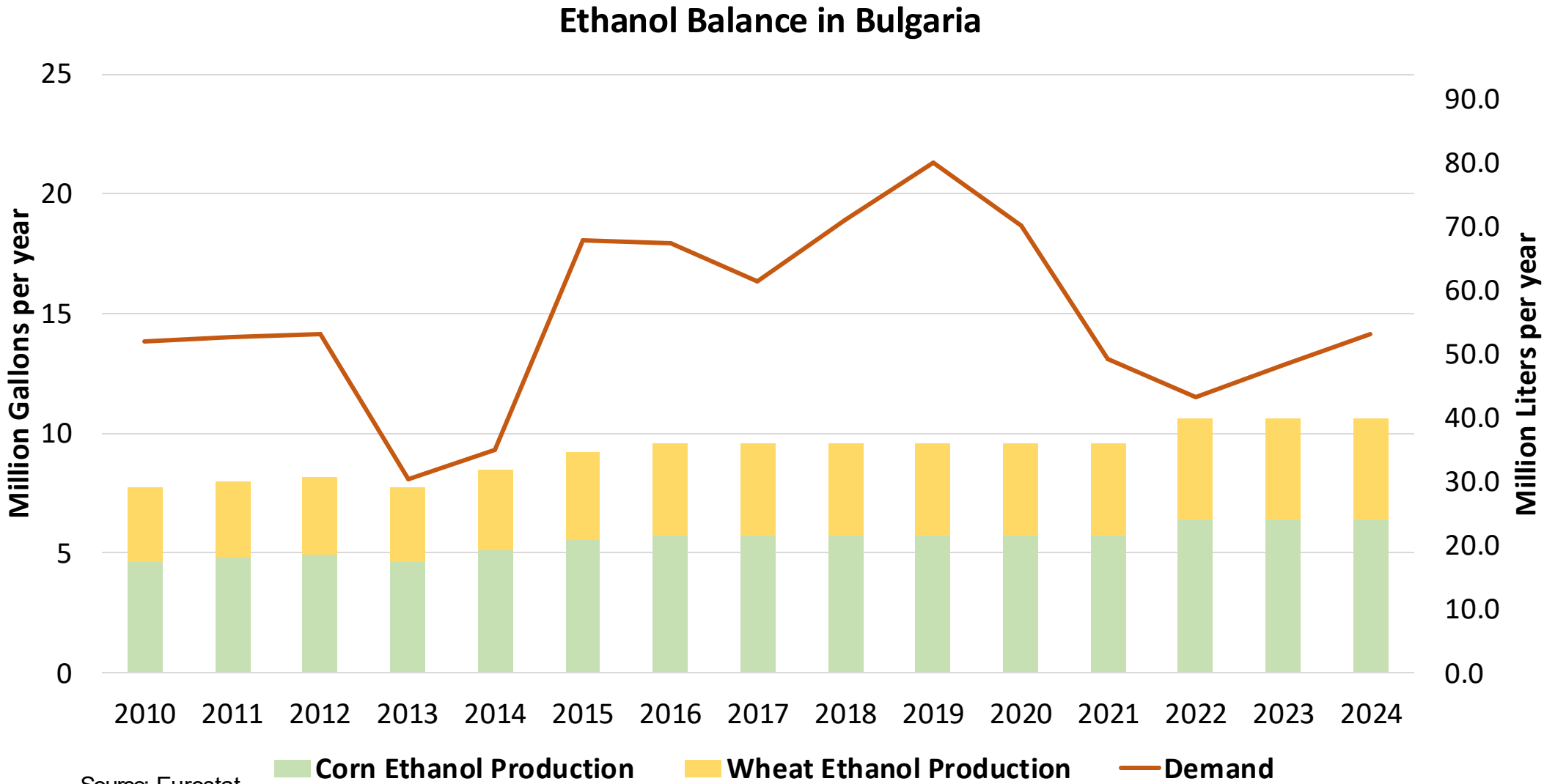
Gasoline Imports in Bulgaria



Gasoline Exports in Bulgaria



Ethanol Balance in Bulgaria



Gasoline Quality in Bulgaria

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

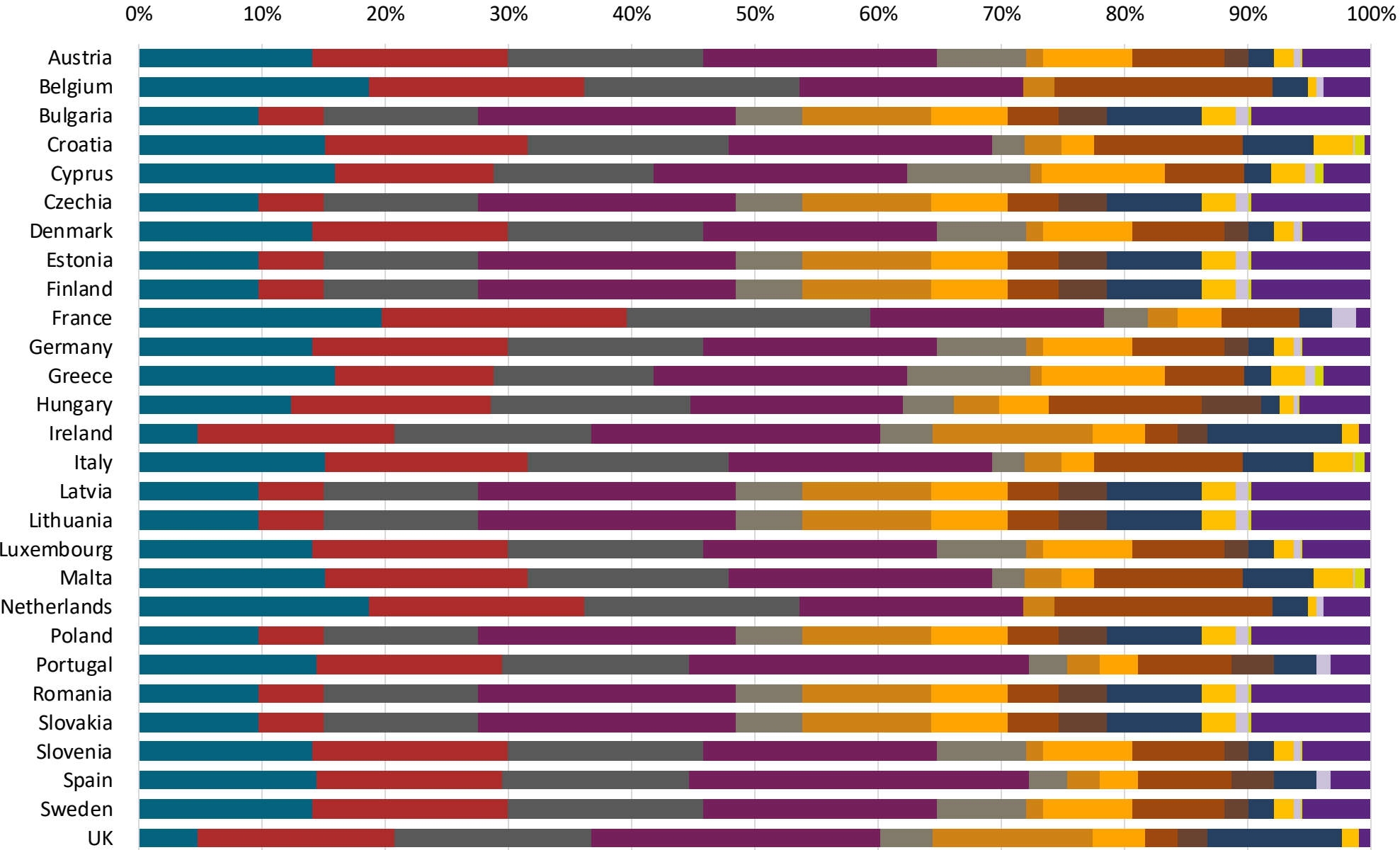
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source: Faro90



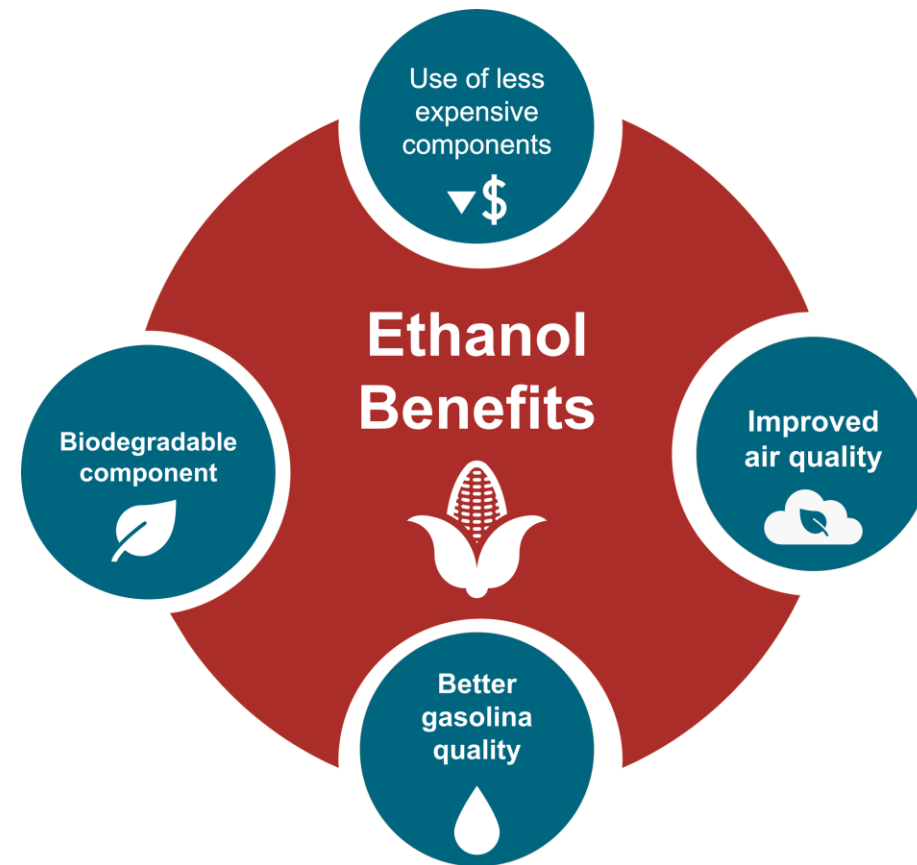
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

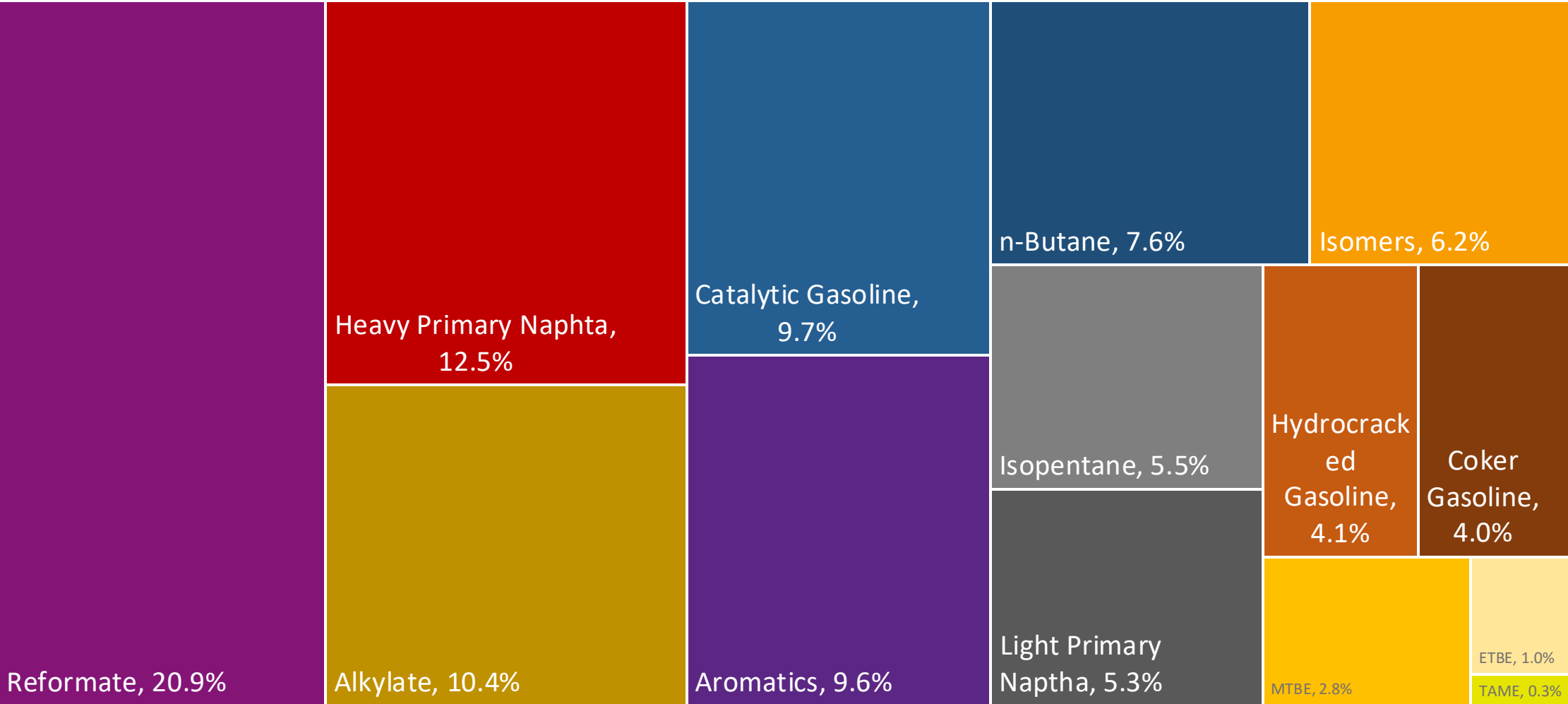
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

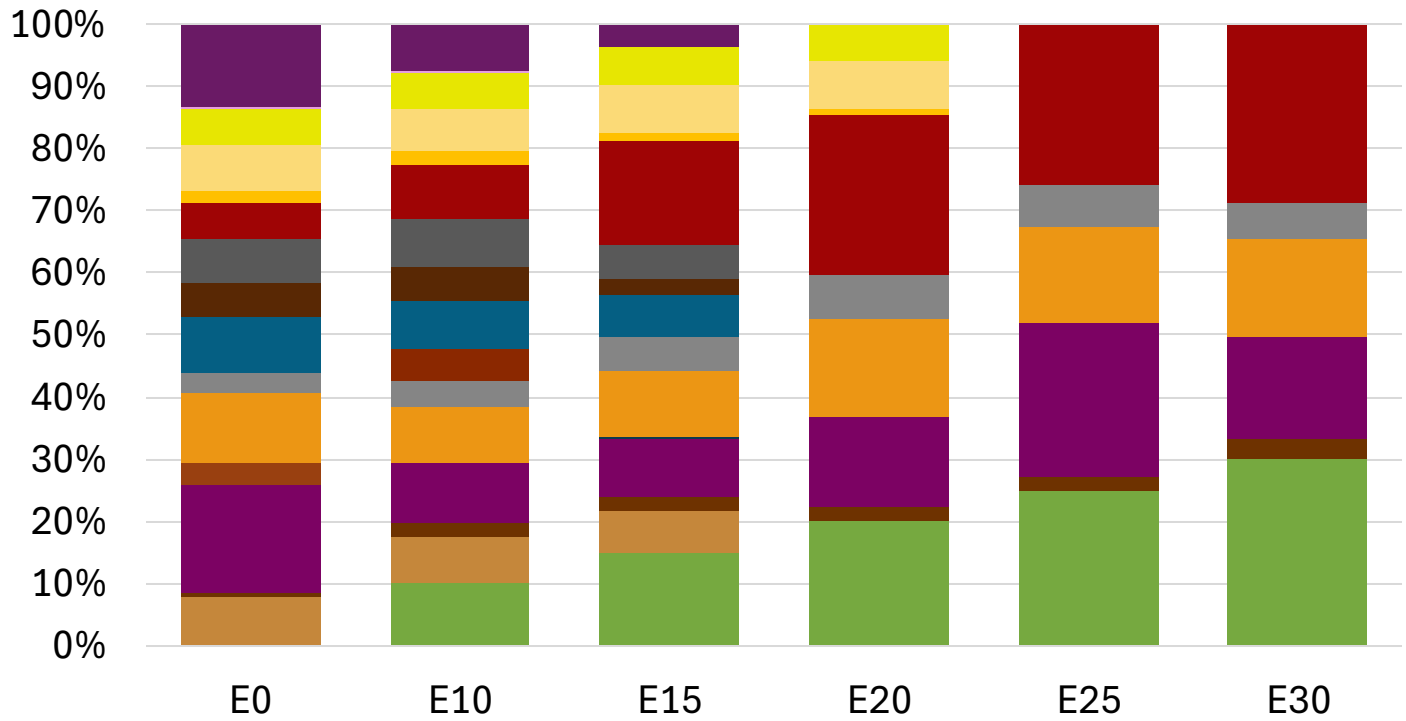
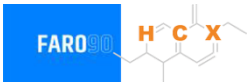
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Bulgaria Available Blending Components



Bulgaria – Gasoline 95 – Constant Octane

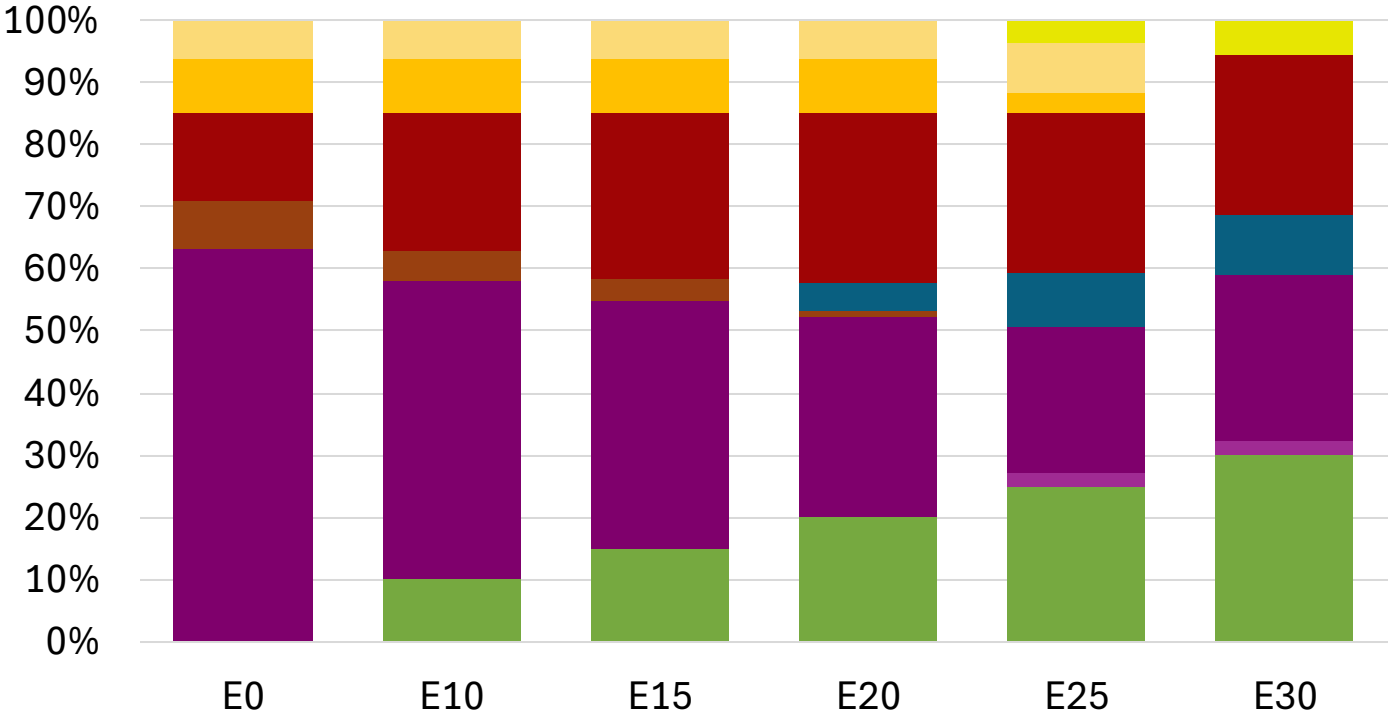
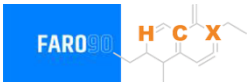


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.294	\$ 2.156	\$ 2.056	\$ 1.941	\$ 1.895	\$ 1.842

Source: Faro90

Bulgaria - Gasoline 98 - Constant Octane

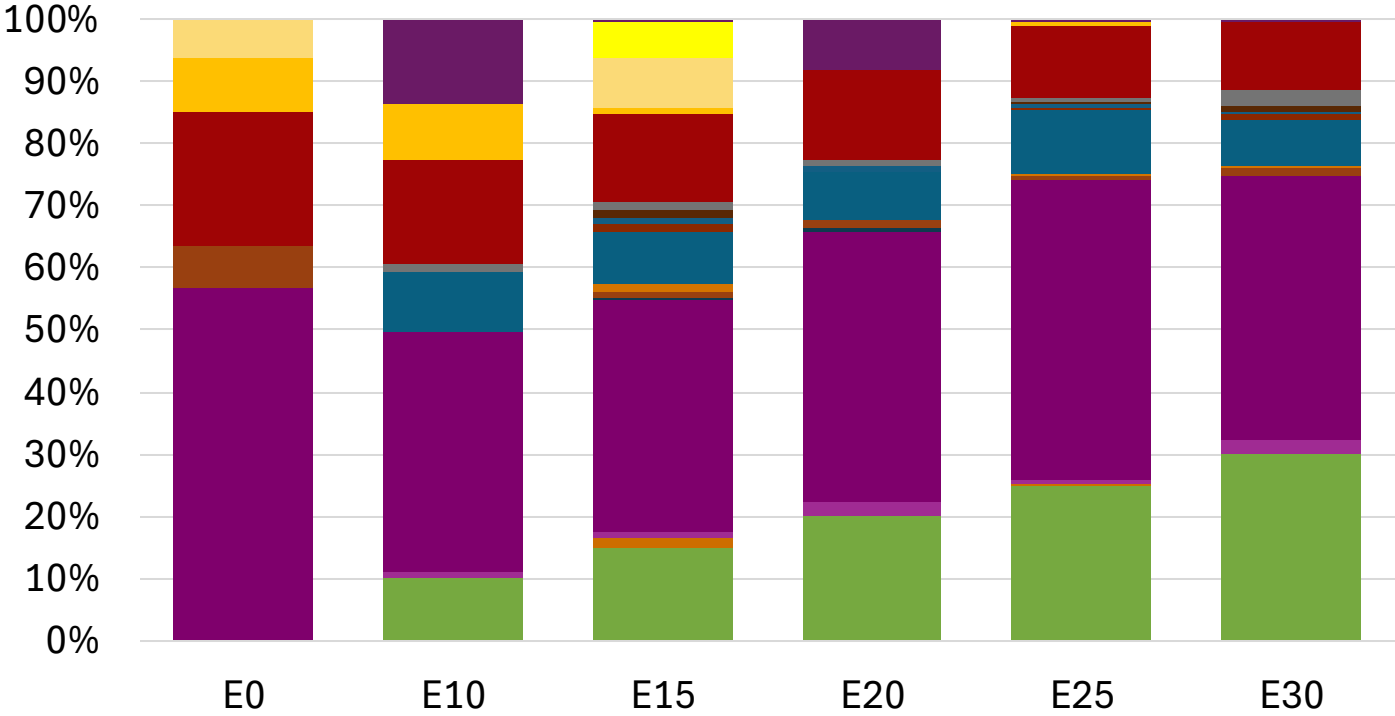
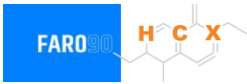


- Ethanol
- Alkylate
- Raffinate
- Reformate
- n-Butane
- Isomerate
- Isopentane
- Hydrocracked Gasoline
- Catalytic Gasoline
- Coker Naphtha
- Light Primary Naphtha
- Heavy Primary Naphtha
- MTBE
- ETBE
- TAME
- Aromatics

Average CIF 2024
Prices

Octane (RON)	98.0					
Price (US\$/gal)	\$ 2.314	\$ 2.197	\$ 2.128	\$ 2.056	\$ 1.991	\$ 1.941

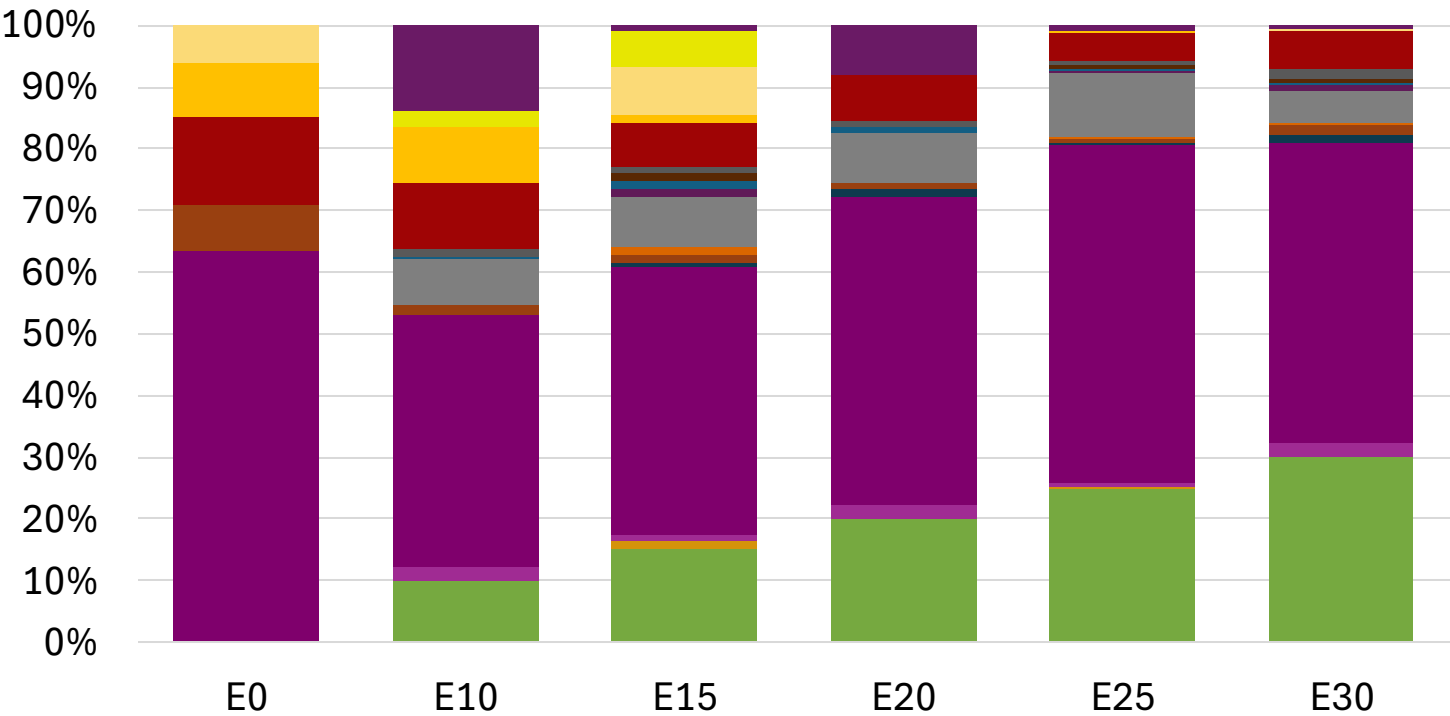
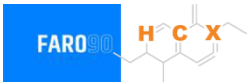
Bulgaria – Gasoline 95 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.182	\$ 2.182	\$ 2.182	\$ 2.182	\$ 2.182	\$ 2.182

Bulgaria – Gasoline 98 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.314	\$ 2.314	\$ 2.314	\$ 2.314	\$ 2.314	\$ 2.314

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

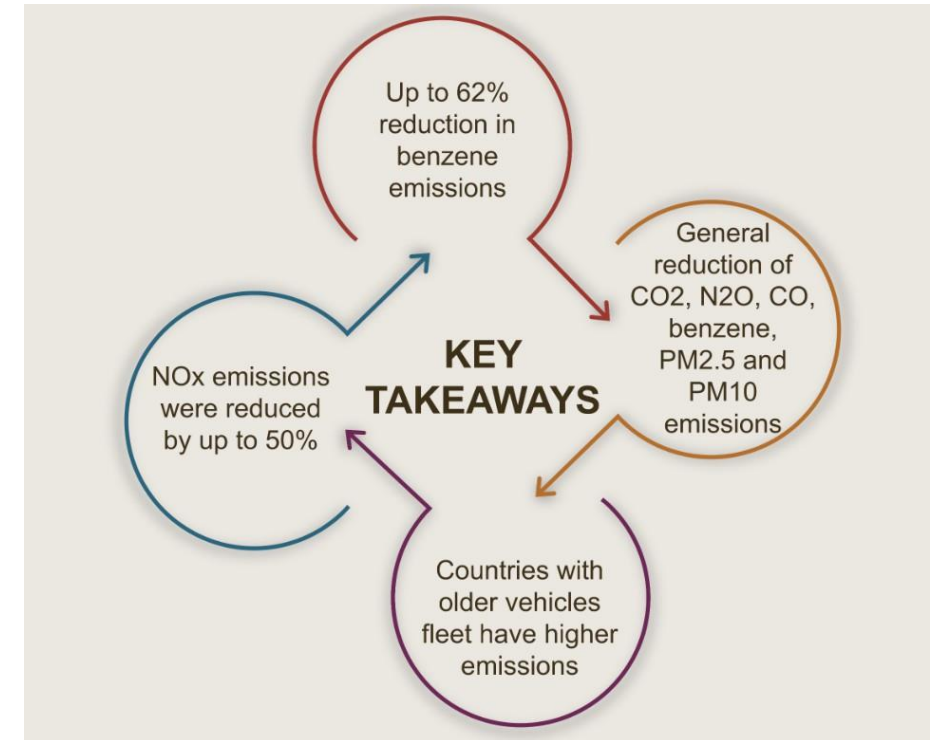
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

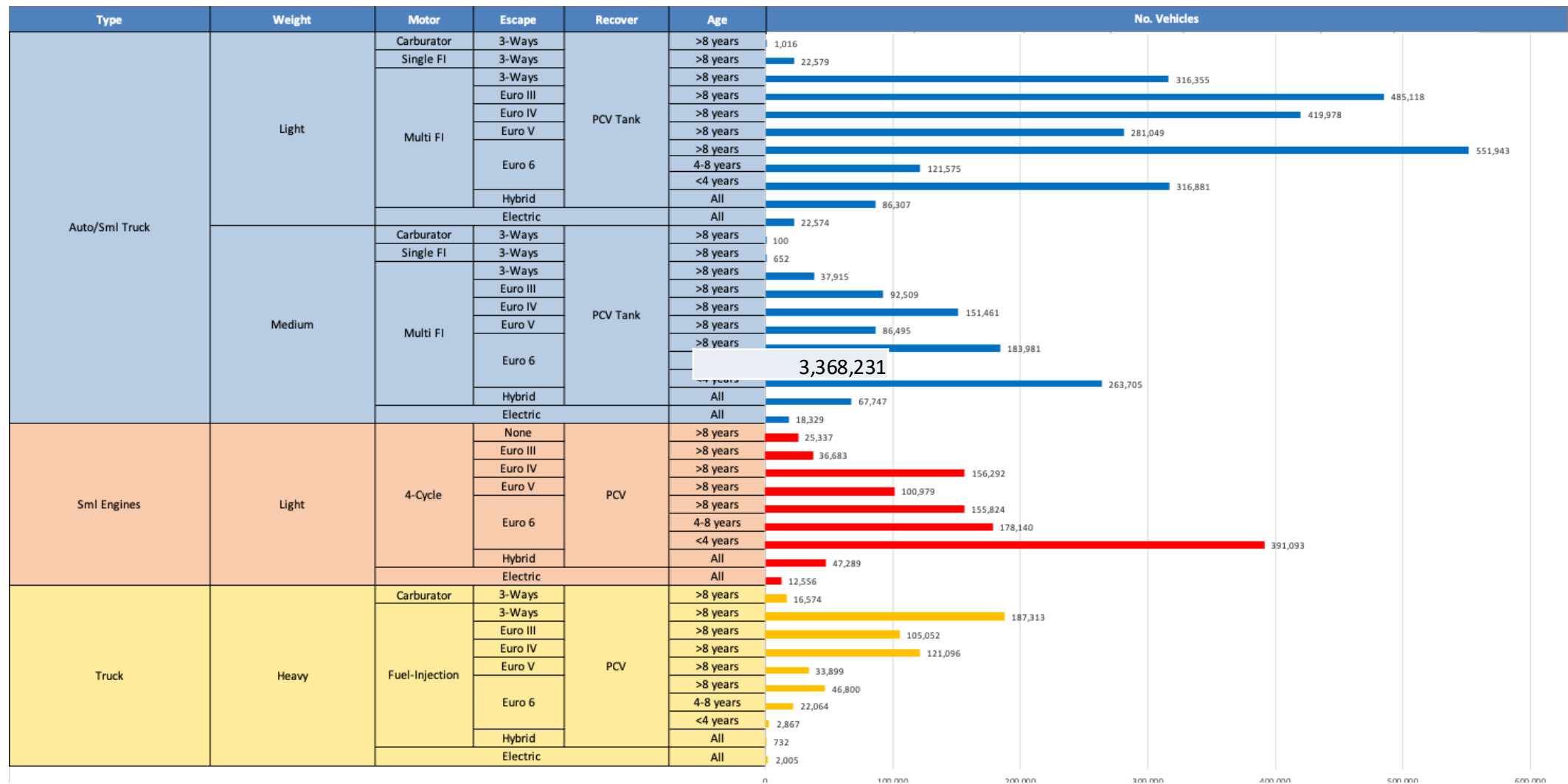
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - Bulgaria

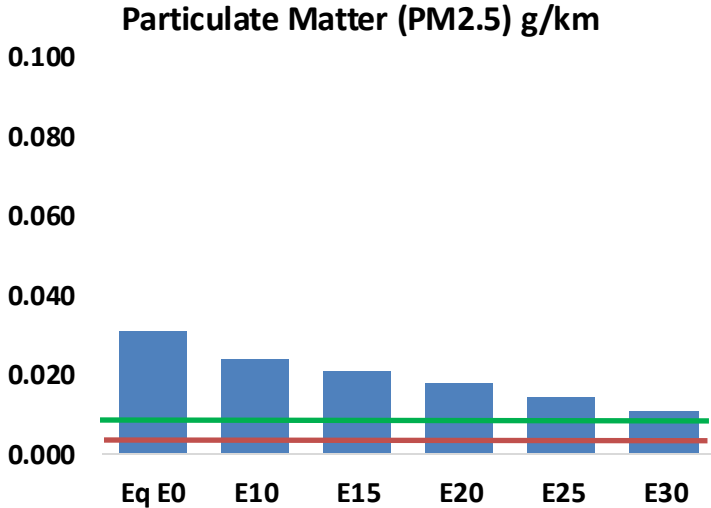
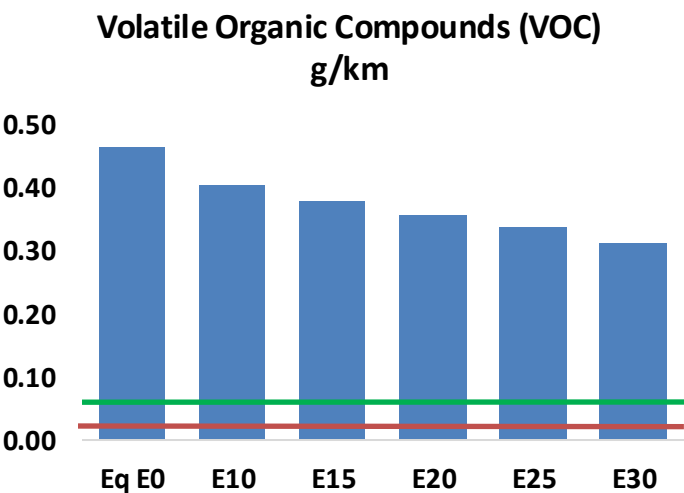
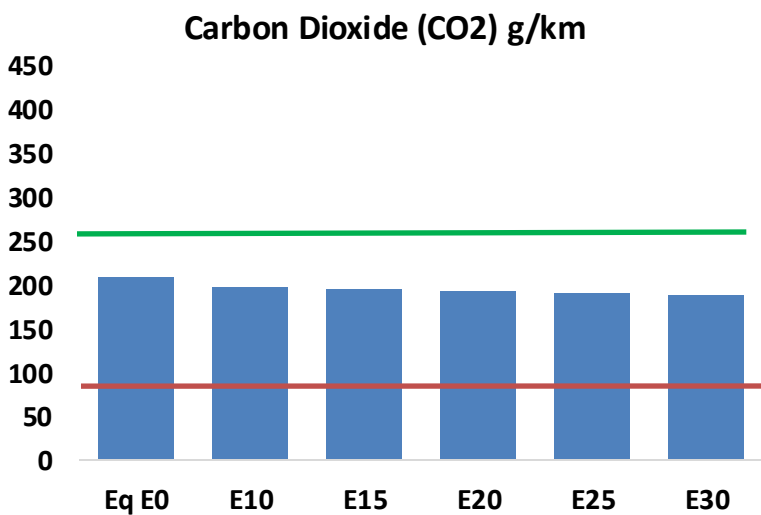
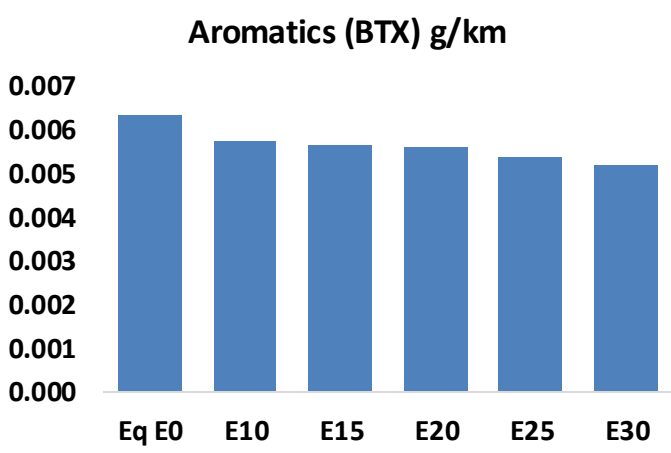
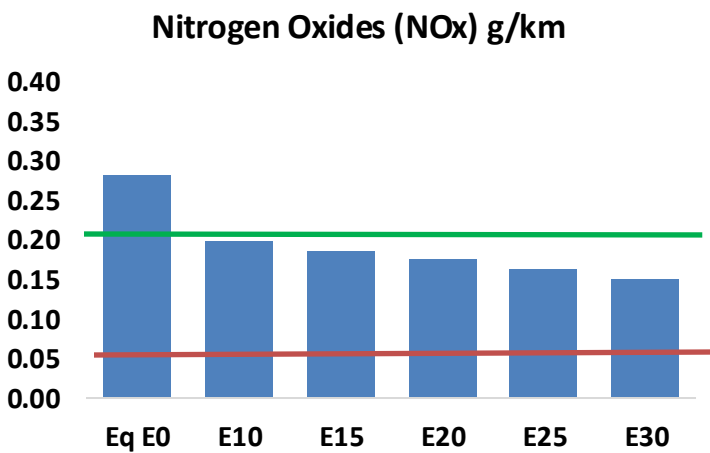
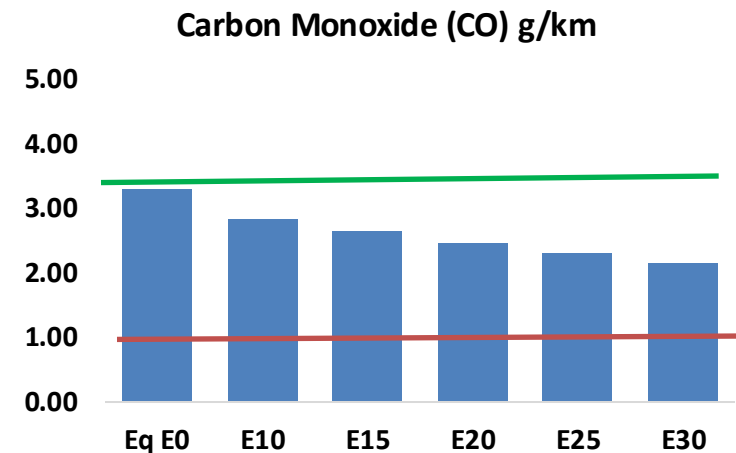
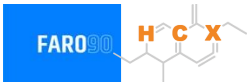


Vehicle Fleet: 5,239,744
Gasoline Vehicle Fleet: 3,368,231

Source: Eurostat, ACEA

Bulgaria – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Bulgaria – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	263,559	244,735	225,910	210,765	195,653	184,072	171,172	-14%	-26%
CO2	16,709,105	16,279,442	15,873,649	15,554,505	15,396,581	15,309,889	15,027,684	-5%	-8%
VOC	37,225	34,766	32,308	30,372	28,460	27,000	25,014	-13%	-24%
NOx	22,636	16,977	15,845	14,934	14,084	13,147	12,156	-30%	-38%
SOx	199	184	169	156	144	131	117	-15%	-28%
NH3	5,918	5,859	5,800	5,828	5,894	5,940	5,978	-2%	0%
N2O	304	302	301	302	309	312	316	-1%	2%
CH4	8,123	8,123	8,123	8,245	8,423	8,578	8,724	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	271	261	252	247	243	240	235	-7%	-10%
Acetaldehyde	701	804	1,180	1,797	2,449	2,798	3,306	68%	249%
Formaldehyde	2,445	2,377	2,771	3,254	3,409	3,772	4,106	13%	39%
Benzene	507	487	461	454	450	431	417	-9%	-11%
PM2.5	1,033	917	801	696	591	480	363	-22%	-43%
PM10	1,437	1,276	1,115	968	822	667	506	-22%	-43%

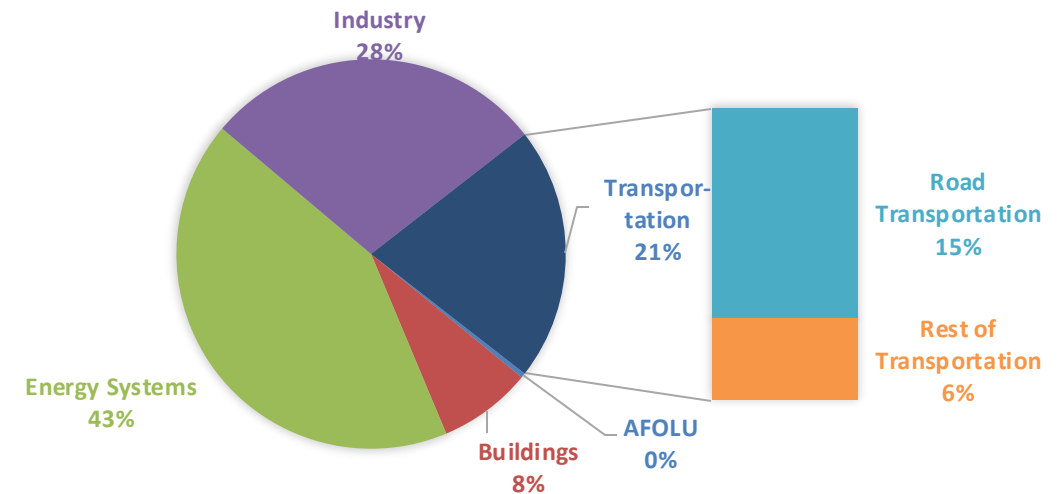
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

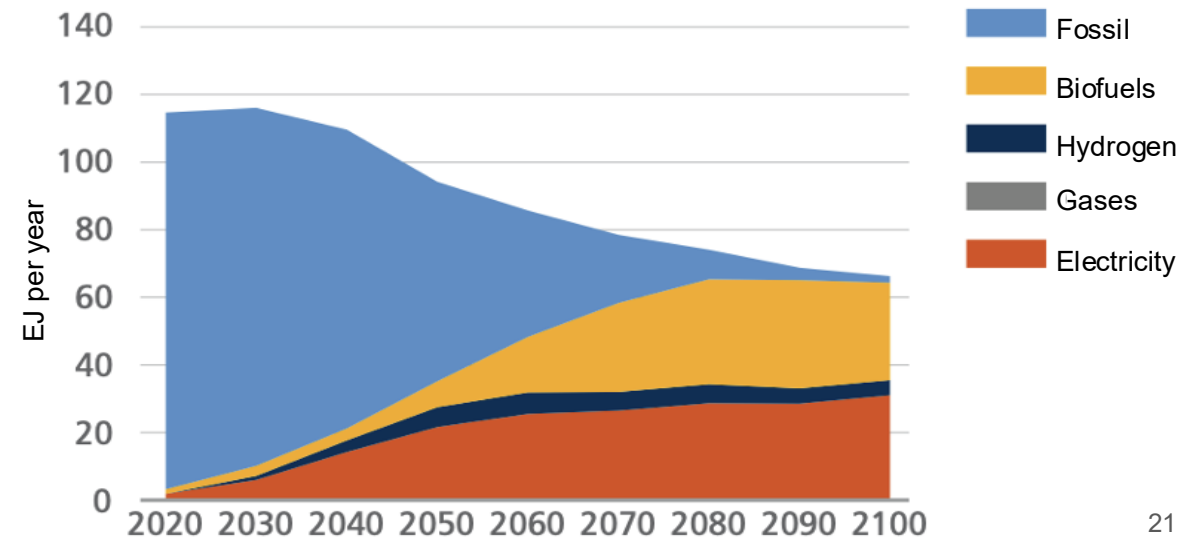
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

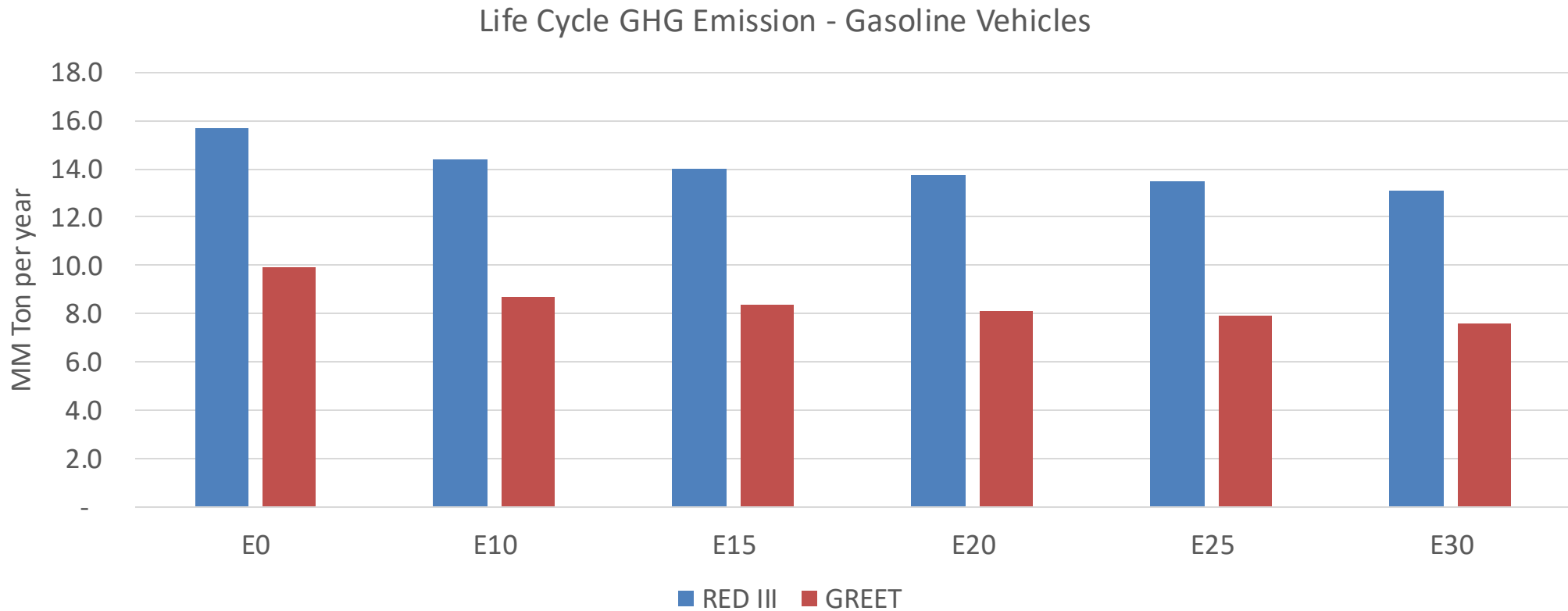
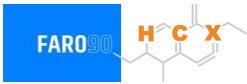
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Bulgaria – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	15.3
Target @ 2035 (MMT/year)	3.1
Delta	12.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.5%	15.7%	18.2%	20.7%	23.9%
RED II/III	0.0%	8.4%	10.6%	12.4%	14.2%	16.4%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	10.2%	12.8%	14.9%	16.9%	19.6%
RED II/III	0.0%	10.8%	13.7%	16.1%	18.3%	21.1%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90